

12

Reporting and diagnostics

This chapter describes the reporting and diagnostics available in Eclipse:

- [Convergence reports](#)
- [Timing reports](#)
- [Total compressibility checks](#)
- [Potential calculations](#)
- [Net carbon production/injection tracking](#)

Convergence reports

x	Eclipse 100
x	Eclipse 300

Both Eclipse 100 and Eclipse 300 can be made to produce reports showing how both the linear and non-linear iterations are proceeding and the methods by which timesteps are selected. The form of this output varies significantly between the two simulators.

Eclipse 100 Report

Mnemonics SUMMARY and NEWTON in the RPTSCHEd keyword control output of reports showing the convergence of the non-linear iterations and a brief summary of the state of the reservoir after each timestep (including non-report times). A sample of the sort of output produced is shown below. The report is copied to the debug file if switch 1 of DEBUG is greater than zero.

```

IT= 0  CNV    CELL      MAT BAL    DPRESS    DSWAT    DSGAS
OIL   .04562( 6, 10, 1)  1.0-002    .00    .00000    .00000
WAT   .00000( 10, 10, 1)  1.6-010    .00    .00000    .00000
GAS   .04780( 1, 1, 1)   6.6-003    .00    .00000    .00000
  LINIT=11 NSCHP=    7 NCHOP=    0 NSTAT1,2,3=    0 219 81 NTRAN= 67

IT= 1  CNV    CELL      MAT BAL    DPRESS    DSWAT    DSGAS
OIL   -.09866( 10, 10, 2)  1.4-004   -170.43   .00013   .01945
WAT   -.00011( 10, 9, 3)  -2.5-006   -233.55   .00006   .00000
GAS   3.80303( 10, 10, 1) -6.8-004   -170.60   .00013   .01982
  LINIT= 8 NSCHP=    0 NCHOP=    0 NSTAT1,2,3=    0 236 64 NTRAN= 85

IT= 2  CNV    CELL      MAT BAL    DPRESS    DSWAT    DSGAS
OIL   -.00744( 10, 10, 2)  -7.4-008   -160.91   .00012   .01986
WAT   .00000( 10, 9, 1)   6.2-011   -222.30   .00016   .02714
GAS   -1.28249( 10, 10, 1) -1.3-008   -162.22   .00012   .03981
  LINIT= 6 NSCHP=    0 NCHOP=    0 NSTAT1,2,3=    0 243 57 NTRAN= 51

IT= 3  CNV    CELL      MAT BAL    DPRESS    DSWAT    DSGAS
OIL   -.00011( 1, 6, 1)    3.8-007   -323.90   .00024   .00579
WAT   -.00000( 10, 10, 1) -5.2-013   -161.05   .00012   .04081
GAS   .00658( 10, 10, 1)  -7.5-007   -161.05   .00012   .04081
  LINIT= 4 NSCHP=    0 NCHOP=    0 NSTAT1,2,3=    0 261 39 NTRAN= 74

IT= 4  CNV    CELL      MAT BAL    DPRESS    DSWAT    DSGAS
OIL   .00001( 8, 2, 1)     2.0-008   -322.26   .00024   .00981
WAT   -.00000( 10, 10, 1) -1.2-016   -161.03   .00012   .04079
GAS   .00004( 2, 4, 1)    -4.1-008   -324.95   .00024   .00508
STEP 19 TIME= 1277.50 DAYS (+182.5 DAYS MAXS 4 ITS) PAV= 5468.8 PSIA
WCT= .000 GOR=    7.53 MSCF/STB WGR= .0000 STB/MSCF

PR.RATE PR.TOTAL IN.RATE IN.TOTAL  MAX SATURN CHANGE  MAX TIME TRNC ERR
OIL 15870. 24540186.    0.    0.   -.041( 10, 10, 1)  .016( 9, 9, 1)
WAT  0.    0.    0.    0.    .000( 1, 1, 3)  -.000( 10, 10, 3)
GAS 119569. 81159402. 100000.127750000.  .041( 10, 10, 1)  -.016( 9, 9, 1)
MAX DP = -330.24( 1, 1, 3) MAX DRS= -.0000( 5, 2, 3)

```

The lines beginning 'IT=' show the progress of the non-linear iteration. The convergence errors for each phase (under CNV) are normalized to appear as saturation errors. The value quoted is the largest in the entire reservoir and the grid block in which it occurs is also reported. The material balance errors for each phase (under MAT BAL) are quoted as a fraction of the total reservoir pore volume. Pressure and saturation changes during the timestep are also reported for the cell with the greatest convergence error. By default, the program continues the iteration until the largest convergence error for any phase in any grid block is less than 0.001, and the largest material balance error is less than 1E-7. These targets may be reset using the TUNING keyword. However such changes are **not** recommended, and can have odd results. It is, for example, quite possible for a loosening of convergence targets to result in **longer** run times. See "Formulation of the equations" for a precise definition of CNV and MAT BAL.

The lines beginning 'LINIT=' give further information about the iteration. LINIT is the number of nested factorization iterations required to solve the linearized equations for the current non-linear iteration. NSCHP is the number of saturation changes which were altered to suppress possible oscillations. NCHOP is the number of times the changes in P , R_s or R_v were reduced (again to increase stability). Any non-zero value of NSCHP or NCHOP has a detrimental effect on material balance errors for the subsequent non-linear iteration. NSTAT1, 2 and 3 are the number of grid blocks in solution states 1, 2 and 3, and NTRAN is the number of inter-state transitions. The significance of the solution states is as follows:

State	Description
State 1	there is no liquid hydrocarbon phase. R_v , the oil-gas ratio in the gas phase, is a variable.
State 2	there are both liquid and gaseous hydrocarbon phases. S_{gas} , the saturation of the gaseous phase, is a variable.
State 3	there is no gaseous hydrocarbon phase. R_s , the gas-oil ratio in the liquid phase, is a variable.

Table 12.1: Solution states

Once the non-linear equations have been solved, Eclipse prints a brief summary of the state of the reservoir at the end of the timestep. On the first line (beginning STEP) are the timestep number, the current time, the length of the timestep just finished, and an argument which indicates the reason why that timestep length was selected. Possible values of this argument are:

INIT	Either the 1st timestep in the run, or a maximum value for the next timestep was set using keyword TUNING or NEXTSTEP.
TRNC	Timestep was set in order to keep estimate of time truncation errors within specified targets.
MINF	Timestep was determined by the minimum permitted ratio between one timestep and the next (default 0.3 - see TUNING). In the PRT file this will usually be followed by the mnemonic (in brackets) that would have been used had this restriction not taken precedence.
MAXF	Timestep was determined by the maximum permitted ratio between one timestep and the next (default 3.0 - see TUNING).
MINS	Timestep is the minimum permitted length (default 0.1 DAYS, see TUNING). In the PRT file this will usually be followed by the mnemonic (in brackets) that would have been used had this restriction not taken precedence.
MAXS	Timestep is the maximum permitted length (default 365 DAYS, see TUNING).
REPT	Timestep was set so as to end at the next report time.
HALF	Timestep is half the interval to the next report time (if timestep would otherwise be larger, but not big enough to reach report time, Eclipse reduces the timestep to half the remaining interval).
CHOP	If the non-linear equations do not converge, a new timestep a tenth the size (see TUNING) is tried (regardless of MINS or MINF limits).
DIFF	Following a CHOP the timestep is expanded cautiously to avoid immediate subsequent chops as the simulation proceeds through a difficult patch. See TUNING (TFDIFF default = 1.25).
THRP	Timestep was determined by the maximum throughput ratio THRUPT (see TUNING).

SATM	Timestep chosen so that the maximum expected saturation change in any cell is DSTARG, set in the IMPES keyword (default = 0.05). This applies only if the IMPES solution method is being used.
MAXP	Timestep chosen so that the maximum expected pressure change in any cell is TRGDPR, set in the TUNING keyword.
PCHP	Timestep chopped because the pressure change was greater than DPMAX, set in the IMPES keyword (default = 200 psi). This applies only if the IMPES solution method is being used.
NETW	Timestep was determined by the network balancing error controls set in items 6-8 in keyword NETBALAN. This applies only to the Network Option.
EMTH	Timestep was set so as to end at the end of the month. This applies only to the GOR Penalty option (see keyword WGORPEN) and the Gas Field Operations Model.
LGRC	Timestep was set in order to keep the estimated fluid-in-place error at the target value TRGFIP set in keyword TUNING. This applies only to the Local Grid Refinement Option.
SURF	Timestep was set in order to keep the estimated surfactant concentration change at the target value TRGSFT set in keyword TUNING. This applies only to the Surfactant Model.
WCYC	Timestep was set to coincide with a well cycling on/off time, or was restricted when a cycling well turned on (see keyword WCYCLE).
SLVR	Timestep in a Reservoir Coupling master run is limited by a report date in a slave reservoir.
SLVC	Timestep in a Reservoir Coupling master run is limited by the allowed flow rate change in a slave.
MAST	Timestep in a Reservoir Coupling slave run is limited to reach the end of the master's timestep, or the timestep in either a slave or master run is limited by the master timestep on the Reservoir Coupling file.
NLTR	Timestep has been chopped due to problems converging non-linear tracer equations. For example, when adsorption is used.
EFF+	Timestep has been selected due to efficiency.
EFF-	Timestep has been selected due to efficiency, but altered slightly to provide new data for the optimization algorithm.
EFFT	Timestep has been selected due to efficiency, but selection has been determined by a non-linear tracer.
ACTN	Timestep has been selected to try and reduce the amount by which an ACTION is violated.
DLYA	Timestep has been selected to enact a delayed action at the appropriate time.
RAIN	Timestep has been selected to coincide with the change of rate at a constant flux aquifer.
MAXW	Timestep is the maximum permitted length following a well modification (see TMAXWC in the TUNING keyword)

On the remainder of the line are the number of Newton iterations required to solve the non-linear equations, the hydrocarbon pore volume weighted field average pressure, and the total water cut, gas-oil ratio and water-gas ratio for field production.

This initial line is followed by a table showing the production and injection rates and totals for each phase, together with the maximum saturation changes and the locations of the grid blocks in which they occur. Also shown are maximum values of the estimated time truncation errors and the locations of the grid blocks in which they occur. Time truncation errors are normalized to appear as saturation errors. This has the side effect of making them appear much more important than they are in fact. In practice, time truncation errors that occur early in a run have a small and rapidly decreasing effect on the later results.

After the table is a single line showing the maximum changes in pressure, R_s and R_v , together with the locations of the grid blocks in which they occur.

Time truncation errors are calculated as

$$\text{TIME TRNC ERR} = dS - \frac{dS_p}{dt_p} dt \quad \text{Eq. 12.1}$$

where

dS is the saturation change during the current timestep, dt , and

dS_p is the saturation change during the previous timestep, dt_p

The maximum value is taken over all cells in the reservoir. A large time truncation error arises when a saturation change is significantly different from that expected.

Output of cells with convergence problems

It is possible to output information to the `RESTART` file providing details of cells that are causing convergence problems. At every non-linear iteration the worst cells have their counter increased by one. Eclipse 100 outputs two sets of mnemonics to the `RESTART` file. The first set provides the worst convergence cells as measured by the magnitude of the residual of every equation, whereas the second set provides the worst cells based on the magnitude of the Newton increments for every variable.

This option is activated by specifying the argument '`CONV=integer`' in keyword `RPTRST`. By default, this 'integer' value is set to 1, so that the single worst cell in the run is focused on. Modifying this 'integer' value alters the number of worst cells.

Depending on the phases present in the model and the options used, different arguments are output to the `RESTART` file. Specifying '`CONV=integer`' in keyword `RPTRST` outputs all relevant arguments for the particular model. The possible arguments are:

Argument	Description
CNV_OIL	Residual of oil equation
CNV_WAT	Residual of water equation
CNV_GAS	Residual of gas equation
CNV_PLY	Residual of polymer equation (<code>POLYMER</code> option)
CNV_SAL	Residual of brine equation (<code>BRINE</code> option when used with polymer)
CNV_SOL	Residual of solvent equation (<code>SOLVENT</code> option)

Argument	Description
CNV_GGI	Residual of Gi Pseudo-compositional Model (GIMODEL option)
CNV_DPRE	Pressure increment
CNV_DWAT	Water saturation increment
CNV_DGAS	Gas saturation increment / R_v increment / R_s increment
CNV_DPLY	Polymer concentration increment
CNV_DSAL	Brine concentration increment
CNV_DSOL	Solvent concentration increment
CNV_DGGI	Dry injection gas parameter (Gi) increment

Note: The active variables for each grid block depend on the solution state (see table 12.1). CNV_DGAS therefore denotes a composite argument reporting the cells with the largest changes in gas saturation, oil-gas ratio in the gas phase, or gas-oil ratio in the liquid phase.

For models with local grid refinements, the worst converging cells are evaluated separately in the global grid and in each local grid. This is because Eclipse 100 by default solves local grids separately using local timestepping. For each grid (global and local) the worst converging cells have their counter increased by 1 after each iteration. The number entered in the CONV argument in [RPTRST](#) is weighted by the ratio of the number of active cells in the local grid to the total number of active cells including global host cells to determine how many local grid cells are output.

Eclipse 300 Report

The simulator outputs information concerning convergence and timesteps to the log file (and echoes this to the PRINT file). A section of this might take the following form:

	Time	Step	GOR	WCT	OPR	WPR	GPR	FPR	WIR	GIR	It
Init;	10.0	10.0	6.46	.026	959.3	26.57	6200	3530.3	340.0	4700	2
SCT ;	25.0	15.0	6.48	.027	955.7	26.76	6200	3524.1	340.0	4700	1
SCT ;	41.0	16.0	6.51	.027	950.9	26.98	6200	3518.1	340.0	4700	1
SCT ;	59.0	18.0	6.54	.027	946.8	27.23	6200	3511.7	340.0	4700	1
SCT ;	80.0	21.0	6.57	.028	942.5	27.51	6200	3504.4	340.0	4700	1
SCT ;	105.0	25.0	6.61	.028	937.6	27.79	6200	3495.8	340.0	4700	1
SCT ;	136.0	31.0	6.65	.029	932.0	28.19	6200	3485.1	340.0	4700	1
SCT ;	175.0	39.0	6.70	.030	925.2	28.73	6200	3471.9	340.0	4700	1
SCT ;	225.0	50.0	6.76	.031	917.0	29.42	6200	3455.2	340.0	4700	1
Max ;	285.0	60.0	6.83	.032	907.0	30.18	6200	3435.6	340.0	4700	1
HRep;	325.0	40.0	6.92	.032	895.8	30.25	6200	3422.9	340.0	4700	1
Rep ;	365.0	40.0	6.97	.033	888.3	30.40	6200	3410.1	340.0	4700	1

The step codes at the left of each line contain the reason for each step. The full set is:

```

ADA : Maximum increase factor for Appleyard timestep selector.
      The number of Newton iterations will be appended to this step code. If this
number is greater than 9 a '+' will be used.
CFL : Stability limited
EFF+ : Timestep chosen by efficient timestep selection method (ZIPPY2)
EnC : Energy Change (thermal)
HRep : Half step to report
Init : Initial step
Max : Maximum step
MaxW : Maximum step after well modification
MDF : Max decrease factor
MIF : Max increase factor

```

```

: Min : Minimum step
: Nite : Number of Iterations
: NVLP : Appleyard timestep selector based on envelope function
: PCT : Pressure change target
: Redo : Redoing the timestep due to a chop
: Redu : Reduced step
: Rep : Step to report
: SAC : Saturation Change (thermal)
: SCT : Solution change target
: SST : Solid solution change target (CO2STORE option)
: Spec : Specified next step
: TEC : Temperature Change (thermal)
: TPT : Throughput target
: TTE : Time Truncation Error
: WAG : Step to WAG change

```

Solution change target (SCT)

Since the primary solution variables are molar densities, it is natural to use these to control the convergence. However, large saturation changes may occur for very small changes in composition, due to the problems of near critical states mentioned in "[Saturation functions](#)", so the use of an effective saturation is preferred. Consider the expression:

$$\Delta S' = \frac{dU^T}{dm_c} \cdot \Delta m_c \quad \text{Eq. 12.2}$$

where

$$u^T = \frac{m_t L}{b_o^m} + \frac{m_t V}{b_g^m} + \frac{m_w}{b_w^m} \quad \text{Eq. 12.3}$$

where

u^T is the fluid volume per unit pore volume

$\Delta S'$ is thus the change in fluid volume per unit pore volume for a given molar density change, that is the effective saturation change for the molar density change.

In all cases the maximum solution change over a timestep is monitored and compared with the specified target. In the `IMPES` and `AIM` cases this is applied in addition to the throughput targets. The simulator will attempt to increase the timestep as long as it is within its current targets, up to the maximum timestep allowed. All these elements of the timestepping algorithm can be modified using the `TSCRT` keyword, and some can be modified using the `TUNING` keyword.

Throughput target (TPT)

The total flow of a fluid (or component) out of a block during a timestep, divided by the amount of that fluid (or component) in the grid block at the start of the timestep defines a throughput ratio.

These throughput ratios usually act as the main timestepping control in `IMPES` and `AIM` modes. (Throughput ratios are closely related to the onset of instabilities in explicit algorithms.) The timesteps are increased or decreased to match a target maximum throughput ratio. This is a maximum taken over all cells and fluids (or components), and is quite conservative.

Both maximum solution change and throughput ratios are subject to limits. The simulator chooses the timestep to attempt to match the target. If the actual value is above this, the step is still accepted, unless the target is exceeded, in which case the timestep is reduced.

Time truncation error (TTE)

A further control on the timestep is provided by the time truncation error (TTE) control. A comparison is made in the change in the solution over a step,

$$\Delta X_a = X^{T+\Delta T} - X^T \quad \text{Eq. 12.4}$$

with the change estimated from the rate of change at the previous step:

$$\Delta X_e = \frac{dX}{dT} \cdot \Delta T \quad \text{Eq. 12.5}$$

where $\frac{dX}{dT}$ is the change per unit time for the previous step.

The time truncation error (TTE) is then estimated from the value of

$$\left| \Delta X_a - \Delta X_e \right| \cdot \frac{dU^T}{dX} \quad \text{Eq. 12.6}$$

where $\frac{dU^T}{dX}$ is used to normalize into dimensionless units of order 1.

Note that if the solution is changing at a constant rate, ΔX_e equals ΔX_a , and there is no TTE limitation.

Minimum step codes

Minimum step codes may start with the two letters Mn followed by an abbreviated version of one of the above - for example MnTP - where the second part indicates what constraint would have applied if the minimum step had not been imposed.

```
MnNC : Minimum timestep, No Convergence
MnSC : Minimum timestep due to Solution change
MnTP : Minimum timestep due to Throughput change
MnPC : Minimum timestep due to Pressure change
MnTT : Minimum timestep due to Time Truncation error
MnTE : Minimum timestep due to Temperature change (thermal)
MnEn : Minimum timestep due to Energy change (thermal)
MnSA : Minimum timestep due to Saturation change (thermal)
```

Maximum decrease factors

Similarly, timestep sizes may be decreased by the maximum decrease allowed for a certain target.

```
MdSC : Maximum decrease due to Solution change
MdTP : Maximum decrease due to Throughput change
MdPC : Maximum decrease due to Pressure change
MdTT : Maximum decrease due to Time Truncation error
MdTE : Maximum decrease due to Temperature change (thermal)
MdEn : Maximum decrease due to Energy change (thermal)
MdSA : Maximum decrease due to Saturation change (thermal)
```

Only lines containing the step code contain the ";" character. This may therefore be used to find such lines, or extract them from the PRT file using a utility such as `grep` on UNIX systems.

Reduction of timestep size may be necessary when the simulator has difficulty converging. The format in the output file when such a reduction is in force is as follows:


```
@--Message at 250.0000 Days 8 Sep 1990
@ Reducing SCT timestep to 15.5000
@ Throughput 1.66718 > limit 1.00000
```

This states that the current timestep, whose size was set to attain a solution change target, was reduced because the cell throughput was greater than the prescribed limit. Reducing a timestep allows the simulator to accurately calculate events that are occurring on a smaller time scale than the original timestep size that was estimated based on the solution change target.

If `RPTPRINT` argument 9 is set to 1, information is also output on the number of cells in each phase/implicitness state. A line is provided for each non-linear iteration, for example:

```
3/OW/GW/W 253 0 63 0 3/OW/GW/W/OG/O/G 5 0 3 0 0 0 0
3/OW/GW/W 240 0 76 0 3/OW/GW/W/OG/O/G 5 0 3 0 0 0 0
```

The first set of numbers are for explicit mobility cells in three-phase, two-phase oil-water, two-phase oil-gas, and water filled states. The second set of numbers are for implicit mobility cells in a similar set of phase states.

Output of cells with convergence problems

It is possible to output information to the RESTART file providing details of cells that are causing convergence problems. At every non-linear iteration, counters are increased by one to indicate the cells that have the largest volume balance residual equations and pressure solution updates. This diagnostic feature can be activated via the 'CONV=*integer*' mnemonic of the `RPTRST` keyword. By default, this 'integer' value is set to 10, which implies that the worst 10 cells in the run are focused on. Modifying this 'integer' value alters the number of cells being monitored. These counter values are output with the arguments `CONV_VBR` and `CONV_PRU` respectively.

Eclipse 300 also outputs the argument `CONV_NEW` which records the number of non-linear iterations required by each cell in order to satisfy the solution change convergence criteria at the last timestep.

These 3D convergence reports can prove useful by enabling problem cells in a run to be pin-pointed. The values are best interpreted relative to one another rather than as absolute numbers.

Example

If 'CONV=40' is used with the `RPTRST` keyword, then the three records `CONV_VBR`, `CONV_PRU` and `CONV_NEW` are output to the RESTART files during the simulation. If a given cell reports a value of 1000 for the `CONV_VBR` data then this implies that the cell has been in the top 40 cells with the largest volume balance residual for 1000 non-linear iterations in the simulation so far. Similarly for the `CONV_PRU` data and the pressure solution update. If a cell reports a value of 3 for the `CONV_NEW` data then this implies that at the previous timestep that particular cell satisfied the solution change convergence criteria on the 3rd non-linear iteration.

Timing reports

x	Eclipse 100
x	Eclipse 300

The argument `CPU` in the `RPTSCHED` keyword (switch number 12) controls the output of timing reports. If output is requested, reports such as that shown below are produced at each report time.

Eclipse 100 timing Reports

```
=====
TIMING REPORTS AT CPU=      558 SECS - ELAPSED=      621 SECS
=====
ZS      1  2  3
        RR      1  2  2
INDX    1  0  0
PR      1  0  0
IR      1  0  0
NNC     1  0  0
XR      1  1  2
JR      1  0  0
YW     12  0  2
TS     11 89 86
        NX     41 88 82
                WC    148  1  1
                        WCDC  148  0  0
                CONV  148  2  2
                PAIR  148  0  0
                LXX   107 68 63
                        WGIN   107  0  0
                        GINV   107  8  8
                        WRRES  107  0  0
                        ITER   107 57 53
                                NORM  980  2  2
                                SRCH  873 34 32
                                WADX  873  0  0
                                ORTH  873 14 13
                                UPDA  873  5  5
                WSOL   107  0
                XUIT  107  2  2
                LC    107 15 14
                API   41  0  0
                TTRC  41  0  0
WWX     11  0  0
MWXR    11  2  2
XW      11  4  4
```

At the top of the report is a statement of the total CPU and elapsed time taken so far by the program. The remainder of the report shows how these times were split between various sections of the program. Each program section is identified by a mnemonic of up to four characters, the most important of which are:

- | | |
|------|---|
| RR | input of the GRID section |
| PR | input of the PROPS section |
| XR | input of the SOLUTION section and equilibration |
| TS | advancing simulator through time |
| NX | solution of non-linear equations by Newton's method |
| WC | setting up well terms in coefficient matrix |
| CONV | checking convergence of non-linear equations |

PAIR pair method (experimental)

MJAC final assembly of Jacobian matrix

LXX solution of the linear equations by Nested Factorization

GINV linear solver initialization

ITER linear solver iteration

NORM linear solver norms (residual max. sums and squares)

SRCH preconditioning and matrix multiply

ORTH orthogonalization

UPDA updating solution and checking linear solver convergence

XUIT state changes, checks on physical reasonableness

LC setting up coefficient matrix (includes property look-up)

API updating API variables

TTRC updating tracer tracking variables

WWX output of well reports

MWXR output of fluid in place reports

XW output of printed tables of pressures and saturations for example

HMO calculation of gradients for the Gradient option

Often, a section is split into smaller sections that are contained entirely within it. For example, NX is split into WC, CONV, MJAC, LXX, XUIT and LC. This structure is shown by the indentation in the above report.

The three numbers given by each section mnemonic are:

- The number of times the section was entered
- The percentage of the total CPU time spent in the section
- The percentage of the total elapsed time spent in each section.

Thus section LXX was entered 107 times; it accounted for 68% of CPU time and 63% of elapsed time. If there were no timestep chops then the number following NX (41) is the number of timesteps so far. Thus the average number of non-linear updates per timestep is $107/41 = 2.61$. Non-linear convergence was checked $107+41 = 148$ times (CONV). The average number of iterations to solve the linear equations was $873/107 = 8.16$. 34% of the CPU time was devoted to solving the preconditioning equations by nested factorization (SRCH). Orthogonalization of each new search direction to the stack of previous search directions (ORTH) required 14% of the CPU time and 13% of the elapsed time. The time spent in ORTH and UPDA may increase **dramatically** if you are forced to use a negative or zero [NSTACK](#). Always make NSTACK positive when possible.

Data inside TS is not recorded when while the CPU mnemonic is absent in [RPTSCHED](#) (or when switch 12 is set to zero). Thus if it is specified just before the final timestep, you will get a report with reliable total

CPU and elapsed times but with an incorrect breakdown of the details. Switching the internal clocks off in this way can save a few percent of the total computing bill.

Eclipse 300 timing reports

```
=====
TIMING REPORTS AT CPU=      52 SECS - ELAPSED=      60 SECS
=====
--XUAIM             1  0.0 %cpu  0.1 %elp
--LSPUT4            3  0.0 %cpu  0.1 %elp
--TS                3 98.7 %cpu 92.7 %elp
----AIM            117  2.1 %cpu  1.8 %elp
-----XUAIM       117  0.0 %cpu  0.0 %elp
-----LSPUT4      117  0.3 %cpu  0.3 %elp
----NLSTEP         117 93.0 %cpu 83.5 %elp
-----DCSP        234 46.2 %cpu 40.3 %elp
-----DCED1A      236 28.2 %cpu 24.5 %elp
-----XSWAPE      236  0.0 %cpu  0.0 %elp
-----DCED1B      236  6.3 %cpu  5.5 %elp
-----WSOLVS      234 15.7 %cpu 14.6 %elp
-----NLRESD      234  2.1 %cpu  1.8 %elp
-----XCNEWD      234 26.5 %cpu 23.5 %elp
-----XCMATV      234  7.2 %cpu  6.3 %elp
-----ELIM        234  2.1 %cpu  1.9 %elp
-----LSSET4      234  0.2 %cpu  0.2 %elp
-----LSOLV       234 13.0 %cpu 11.4 %elp
-----LWARP       234 13.0 %cpu 11.4 %elp
-----LGFAC       234  2.2 %cpu  2.1 %elp
-----LGPGMR      234 10.2 %cpu  8.8 %elp
-----LGFBS      1877  6.0 %cpu  5.2 %elp
-----LGMLT      1877  2.1 %cpu  1.8 %elp
-----BACKSU      234  1.8 %cpu  1.6 %elp
-----XLIMIT      234  0.2 %cpu  0.2 %elp
-----NLUPDT      234  0.1 %cpu  0.2 %elp
----PREDA         585  0.0 %cpu  0.0 %elp
----PSUMR         21  0.0 %cpu  0.0 %elp
=====
```

Again, as with the Eclipse 100 timing reports, the top of the report contains a statement which details the total CPU and elapsed time taken so far by the program. The remainder of the report shows how these times were split between various sections of the program. Each program section is identified by a mnemonic of up to six characters, the most important of which are:

AIM	Adaptive IMplicit method
AJGRAD	Linear solution for adjoint vector
DCSP	Main domain calculation
GHMCAL	Main routine for gradient calculation
GSOLVS	Supervises the gradient calculations at the end of a timestep
LSOLV	WARP linear solver
LWARP	WARP linear solver
NLCONV	Check for convergence of the non-linear equations solved using Newton's method
NLRESD	Set up the residual for the Newton system
NLSTEP	Solution of the non-linear equations by Newton's method

NLTRAC Solution of the tracer tracking variables

NLUPDT Update the solution variables during a Newton iteration

TS Advances the simulator through time to the next report step

WSOLVS Solve the production and injection well system

XCFLOW Calculate the flows

XCMATV Set up the Jacobian matrix

XLIMIT Limit the solution changes during a Newton iteration

XUAIM Set up the Adaptive IMplicit (AIM) indicator array

Total compressibility checks

x	Eclipse 100
x	Eclipse 300

In black oil models, Eclipse 100 and Eclipse 300 check for positive compressibility of each single reservoir fluid as the PVT data is read (the formation volume factor must be a monotonically decreasing function of pressure assuming all other variables are held fixed).

However, Eclipse also checks that oil-gas mixtures have a positive total compressibility even when there is mass transfer between the two phases. For example, it is well known that oil sometimes swells when the pressure increases, apparently in violation of physical principles. This swelling occurs because gas is dissolving in the oil. Provided that the reduction in the volume of gas is greater than the increase in the volume of oil, the total volume of the oil-gas mixture reduces as the pressure increases, and the paradox is resolved.

It is possible to calculate a value for the total compressibility of a hydrocarbon system where mass exchange may occur between the reservoir liquid and vapor phases. Consider a volume V_g of separator gas and a volume V_o of stock tank oil at surface conditions. Compress the mixture to pressure P , volume V and assume that a two-phase equilibrium state is formed. The material balance equations for the oil and gas components are

$$V_g = \frac{V \cdot S_g}{B_g} + \frac{V \cdot R_s \cdot S_o}{B_o} \quad \text{Eq. 12.7}$$

$$V_o = \frac{V \cdot S_o}{B_o} + \frac{V \cdot R_v \cdot S_g}{B_g} \quad \text{Eq. 12.8}$$

Equations 12.7 and 12.8 allow the mixture volume V and gas saturation S_g to be computed. If the pressure of the system is increased, the volume occupied by the liquid-vapor system must decrease for physically well-defined fluids. The total compressibility of the system, $C_t = -(1/V)(dV/dP)$, can be shown to be given by the following expression

$$C_t = \frac{S_g}{B_g} \left(-\frac{dB_g}{dP} + \frac{dR_v}{dP} \cdot \frac{B_o - R_s B_g}{1 - R_s R_v} \right) + \frac{S_o}{B_o} \left(-\frac{dB_o}{dP} + \frac{dR_s}{dP} \cdot \frac{B_g - R_v B_o}{1 - R_s R_v} \right) \quad \text{Eq. 12.9}$$

where the pressure derivatives are taken along the saturation curve.

This can also be written as

$$C_t = S_g C_{t,gas} + S_o C_{t,oil} \quad \text{Eq. 12.10}$$

which splits the total compressibility into gas and oil dependent terms.

To derive this expression, we assume without loss of generality that

$$S_o + S_g = 1 \quad \text{Eq. 12.11}$$

and note that

$$dV_o/dp = dV_g/dp = 0 \quad \text{Eq. 12.12}$$

since the volumes of stock tank oil and separator gas are fixed.

Solving for S_o and S_g gives

$$S_o = \frac{R_v V_g - V_o}{V} \cdot \frac{B_o}{R_s R_v - 1} \quad \text{Eq. 12.13}$$

and

$$S_g = \frac{R_s V_o - V_g}{V} \cdot \frac{B_g}{R_s R_v - 1} \quad \text{Eq. 12.14}$$

And since $S_o + S_g = 1$ we have

$$V = \frac{1}{(R_s R_v - 1)} \cdot \left(V_o (R_s B_g - B_o) + V_g (R_v B_o - B_g) \right) \quad \text{Eq. 12.15}$$

Hence

$$\begin{aligned} C_t &= -\frac{1}{V} \cdot \frac{dV}{dp} \\ &= -\frac{(R_s dR_v/dp + R_v dR_s/dp)}{(R_s R_v - 1)} \\ &\quad + \frac{\left\{ V_o \left[R_s \frac{dB_g}{dp} + B_g \frac{dR_s}{dp} - \frac{dB_o}{dp} \right] + V_g \left[R_v \frac{dB_o}{dp} + B_o \frac{dR_v}{dp} - \frac{dB_g}{dp} \right] \right\}}{V_o (R_s B_g - B_o) + V_g (R_v B_o - B_g)} \end{aligned} \quad \text{Eq. 12.16}$$

which after some manipulation, and substitution for V_o and V_g using the above relations, leads to the desired expression for C_t .

Some simple limiting cases may be derived from equation 12.9.

- If $R_s = R_v = 0$ then

$$C_t = -\frac{S_g}{B_g} \cdot \frac{dB_g}{dP} - \frac{S_o}{B_o} \cdot \frac{dB_o}{dP} > 0 \quad \text{Eq. 12.17}$$

since the formation volume factor pressure derivatives are < 0 .

- If $R_s > 0$ and $R_v = 0$ then

$$C_t = -\frac{S_g}{B_g} \cdot \frac{dB_g}{dP} - \frac{S_o}{B_o} \cdot \left(\frac{dB_o}{dP} - B_g \frac{dR_s}{dP} \right) \quad \text{Eq. 12.18}$$

so that $C_t < 0$ if $dB_o/dP > B_g dR_s/dP$ and S_g is sufficiently small.

Eclipse checks the total compressibility of the hydrocarbon PVT data as it is read from the input data file. For each PVT region in the simulation grid a pressure range is selected from the corresponding oil and gas PVT data tables to span the complete range of pressure data in the two tables. This pressure range is then subdivided into 30 equally spaced pressure nodes for the evaluation of the total hydrocarbon compressibility using equation 12.9.

The sample pressure range is the maximum of the oil pressure (maximum bubble point pressure in [PVTO](#) or maximum pressure specified in [PVDO](#)) and the gas pressure (maximum dew point pressure specified in [PVTG](#) or maximum gas pressure specified in [PVDG](#)).

Eclipse 100

The maximum sample pressure at which total compressibility is checked can be overridden by choosing the second item in the [PMAx](#) keyword. This can be used to check for negative compressibility in extrapolated regions.

At each pressure node two limiting compressibility values are calculated corresponding to $S_g = 0$ and $S_g = 1$ in equation 12.9. If either or both values should be negative, a warning message will be issued indicating the offending pressure value and the gas saturation range for which the compressibility is negative. It should then be possible using equation 12.9 to determine how to change the saturated gas/oil PVT data to ensure the total compressibility remains positive for all pressures.

If the range of sample pressures extends above the maximum bubble point specified in the PVTG table or above the maximum dew point specified in the PVTG table, then Eclipse is forced to extrapolate above the highest entered bubble point or dew point. This extrapolation is linear in $1/B_o$ and $1/(B_o\mu_o)$ etc. (see the PVTG keyword). In this case, it is not unlikely that negative compressibilities could occur as a result of extrapolation.

In some cases, if the pressure is extrapolated severely, then the linearly-extrapolated formation volume factors may become unphysical. It is recommended that the highest bubble point nodes in the PVTG table are constructed so avoid extrapolations above the highest entered R_s in the PVTG table during the simulation (similarly, in a run with vaporized oil present, it is recommended that the highest dew point nodes in the PVTG table are constructed to avoid extrapolations above the maximum entered P_{dew} during the simulation).

Eclipse 100

The [EXTRAPMS](#) keyword can be used to instruct Eclipse 100 to report PVT table extrapolations during the simulation.

Eclipse 100

If the 21st switch in the [DEBUG](#) keyword is set > 0 , Eclipse 100 reports the total gas and oil compressibilities $C_{t,gas}$ and $C_{t,oil}$ as well as $R_{s,sat}$, $B_{o,sat}$ and $R_{v,sat}$, $B_{g,sat}$ for the sample range of pressures to the debug file (extension. `DBG`) in tabular form. Whenever any negative compressibilities are encountered in the data checking, this debug output is automatically switched on. This may then be used to diagnose any negative compressibilities reported by Eclipse 100.

Note: The effect of a negative total compressibility is usually to cause the simulator to experience numerical difficulties (convergence failures and/or erratic changes in the solution) in regions where the total compressibility is negative.

Potential calculations

Phase potentials

x	Eclipse 100
x	Eclipse 300

Eclipse 100

The phase potentials are the oil, water and gas phase pressures corrected to a datum depth. Eclipse can optionally calculate them for output. They may be written to the PRINT file, the SUMMARY file and the RESTART file.

The phase potential of a cell is calculated by correcting the appropriate phase pressure to the datum depth using the average phase density (at reservoir conditions) in the PVT region containing the cell,

$$\begin{aligned}\Phi_{\text{oil}} &= P_{\text{oil}} - \rho_o g (D - \text{DATUM}) \\ \Phi_{\text{wat}} &= P_{\text{wat}} - \rho_w g (D - \text{DATUM}) \\ \Phi_{\text{gas}} &= P_{\text{gas}} - \rho_g g (D - \text{DATUM})\end{aligned}\quad \text{Eq. 12.19}$$

where

ρ_o = PVT region average oil density

ρ_w = PVT region average water density

ρ_g = PVT region average gas density

D = depth of the cell

DATUM = datum depth

g = acceleration due to gravity

Eclipse 300

The phase potential of a cell is calculated by using the block density rather than the PVT region average density.

The datum depth is either supplied using the DATUM keyword, or is taken to be the datum depth specified for the equilibration region 1, in the EQUIL keyword. If both the DATUM and EQUIL keywords are absent, the datum depth is set to zero. The datum depth can be provided for each fluid-in-place region using the keywords DATUMR or DATUMRX, in which case the value specified in any DATUM keyword is ignored.

The region and field average potentials are the phase pore volume weighted averages of the cell potentials within a region.

Hence, for a region or field, the average potential of phase p with saturation S_p is given by

$$\bar{\Phi}_p = \frac{\Sigma \Phi_p \cdot PV \cdot S_p}{\Sigma PV \cdot S_p} \quad \text{Eq. 12.20}$$

Eclipse 100

Setting OPTIONS keyword [switch 31](#) non-zero and divisible by two, weights the potential by pore volume rather than phase pore volume.

Eclipse 300

Setting OPTIONS3 keyword [switch 86](#) to 1 weights the potential by hydrocarbon pore volume rather than by phase pore volume.

The PVT region average density is always weighted by phase saturation, so that

$$\bar{\rho}_p = \frac{\sum PV \cdot S_p \cdot \rho_p}{\sum PV \cdot S_p} \quad \text{Eq. 12.21}$$

where the sum is over a PVT region, and PV is the reference pore volume. When the phase saturation is zero within a PVT region (for example, oil phase saturation for a gas condensate above the dew point pressure), a pore volume weight is used instead to obtain the average density which is only used to report what the potential would be if the phase occupied the PVT region.

Initial contact corrected potential (Eclipse 100)

Eclipse 100 also calculates a representative potential, where the pressure is corrected to the datum depth using the water density below the initial WOC, the gas density above the initial GOC, and the oil density between the two initial contacts.

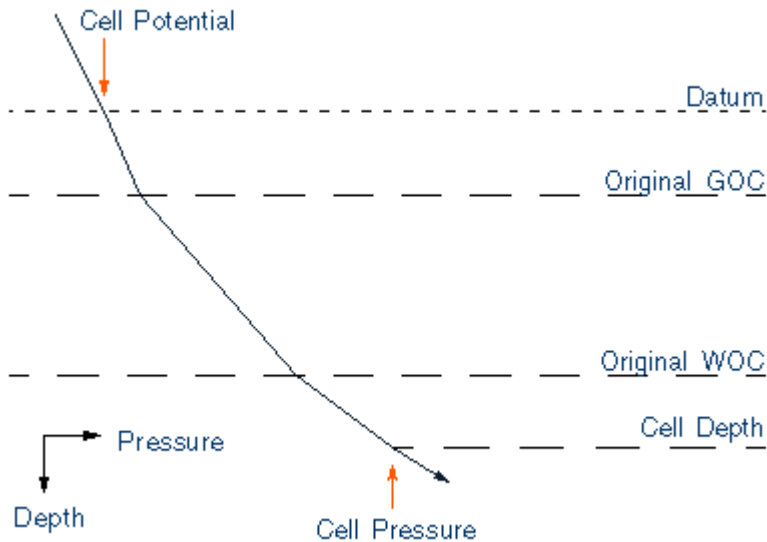


Figure 12.1. Calculating the initial contact corrected potential

This representative potential can be thought of as the potential of the phase occupying the region at initial equilibrium conditions.

Output of potentials

The potentials can be output using the following keywords.

Eclipse 100

To the PRINT file:

```
RPTSOL
'POTO' 'POTW' 'POTG' 'POTC' /
RPTSCHED
'POTO' 'POTW' 'POTG' 'POTC' /
```

Eclipse 300

To the PRINT file:

```
RPTSOL
'OILPOT' 'GASPOT' 'WATPOT' /
RPTSCHED
'OILPOT' 'GASPOT' 'WATPOT' /
```

To the SUMMARY file:

FPP0	RPP0	BPP0	Oil Potential
FPPW	RPPW	BPPW	Water Potential
FPPG	RPPG	BPPG	Gas Potential

Eclipse 100

Again, for the SUMMARY file:

FPPC	RPPC	BPPC	Initial contact corrected potential
------	------	------	-------------------------------------

Eclipse 100

To the RESTART file:

```
RPTRST  
'POT' /
```

Well potentials

Well potentials are flow rates rather than pressures. They represent the production or injection rate that a well would achieve in the absence of any rate constraints, at the current grid block conditions. The acting constraint for a well's potential is thus either its BHP limit or its THP limit, whichever is the more restrictive. If a well has a drawdown limit, that constraint may optionally be included in its potential calculation by setting the flag in item 4 of keyword [WELDRAW](#).

Eclipse 100

There is an option to ignore THP limits when calculating the potential, which may be useful if the potentials are not used for anything important and the potential flows would extrapolate the VFP tables (see [OPTIONS](#) keyword [item 40](#)).

Note that if a well has no VFP table and its BHP limit is defaulted, its potential will not be particularly meaningful.

The calculated well potential may be somewhat optimistic. This is because the grid block conditions at which the potential is calculated are not changed to take account of the well's higher flow rate. In reality, increasing a well's production from its actual rate to its potential would cause the pressure in its connecting grid blocks to decrease, thus decreasing its potential. The effect may be insignificant for wells completed in large grid blocks, but it could be appreciable for wells completed in small grid blocks, especially when using local grid refinement. The smaller the connecting grid blocks are, the more optimistic will be the potential.

On a similar note, the potential of a well belonging to a network (see "[Network Option](#)") may also be optimistic for another reason. The THP limit at which the well's potential is calculated is the same as for its normal flowing rate. In reality, increasing a well's production from its actual rate to its potential would cause its THP limit to increase due to the greater pressure losses in the network, thus decreasing its potential.

Eclipse uses well potentials in the following calculations and facilities:

- To set guide rates for group control, for wells that have not been given specific guide rates in keyword [WGRUPCON](#). In Eclipse 100, the well and group guide rates can be set according to a function of their potentials (see keyword [GUIDERAT](#)).
- When using prioritization group control, well priorities can be set according to a function of their potentials (see keyword [PRIORITY](#)).

Eclipse 100

- Wells can be closed automatically when their potential production or injection rates fall below an economic limit (see keywords [WECON](#) and [WECONINJ](#)).
- Eclipse 100* A well may be opened from the drilling queue when required to maintain a group's production or injection potential above a specified value (see keyword [GDRILPOT](#)).
- In a Prioritized Drilling Queue, well drilling priorities can be set according to a function of their potentials (see keyword [DRILPRI](#)).
- Eclipse 100* Well potentials may be written to the PRINT file, in place of their PI, by setting a negative value for the drainage radius in item 7 of keyword [WEL SPECS](#). Well, connection and group potentials can also be written to the SUMMARY file, using the keywords:

Field	Group	Well	Connection	Description
FOPP	GOPP	WOPP	COPP	Oil Potential Production rate
FOPI	GOPI	WOPI	COPI	Oil Potential Injection rate
FWPP	GWPP	WWPP	CWPP	Water Potential Production rate
FWPI	GWPI	WWPI	CWPI	Water Potential Injection rate
FGPP	GGPP	WGPP	CGPP	Gas Potential Production rate
FGPI	GGPI	WGPI	CGPI	Gas Potential Injection rate

In Eclipse 300 well potentials can be written to the SUMMARY file, using the keywords:

Well	Description
WOPP	Oil Potential Production rate
WWPP	Water Potential Production rate
WWIP	Water Potential Injection rate
WGPP	Gas Potential Production rate
WGIP	Gas Potential Injection rate

The field and group potentials are the sum of their subordinate wells' potentials. The connection potentials are the individual connection flow rates when the well is operating at its potential.

Net carbon production/injection tracking

Eclipse provides a mechanism for tracking the net amount of carbon produced or injected during a simulation.

This is purely a mechanism for tracking the movement of carbon into or out of the reservoir. It is not a full carbon accounting mechanism because it does not account for the carbon that is consumed or generated by the operation of the reservoir (for example, the drilling of wells, the operation of plants and wells, or the transportation of resources and fluids) because the required information for such a calculation is not typically available to the simulator.

For the purposes of net carbon production/injection tracking, **only the movement of components that contain carbon into or out of the reservoir system is of interest**. The phase in which these components exist is unimportant because, regardless of the phase, the component always contains the same amount of carbon. The movement or location of the carbon within the reservoir system is also unimportant. For example, it is not required to track:

- Flows from one region of the reservoir to another (either through the grid cells or through crossflow in wells) because this is just a redistribution of carbon *within* the reservoir system.
- Flow into, or out of, aquifers because this is all considered *part of* the reservoir system.
- Adsorption of components onto rocks in the reservoir (for example, with polymer or surfactant modeling) because this occurs *within* the reservoir system.

The wells are the only entry and exit points for carbon in the reservoir system. Therefore, net carbon production/injection is calculated as the difference between the total moles of carbon produced at the well heads and the total moles of carbon injected at the well heads (see figure 12.2 where the well head flows are the only ones that cross the boundary of the reservoir system).

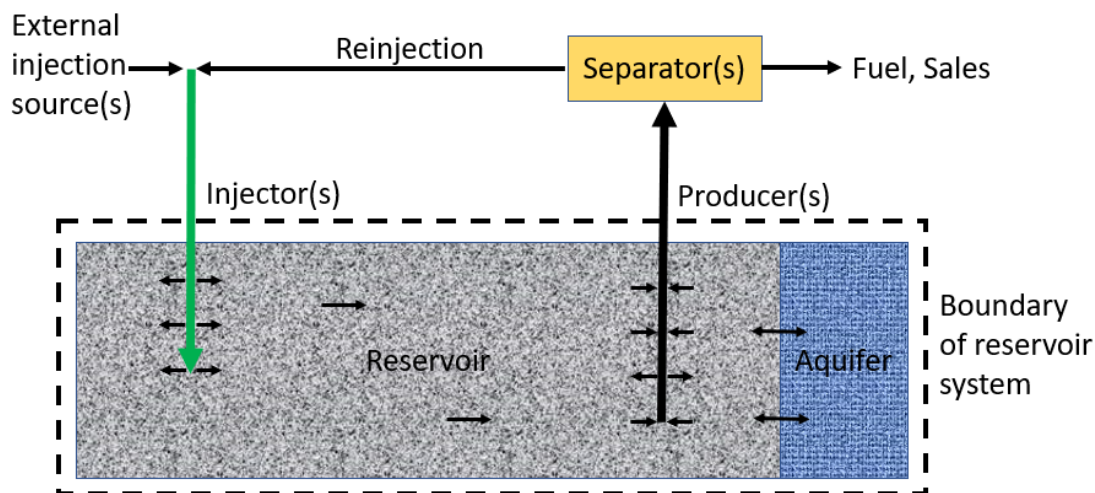


Figure 12.2. The movement of carbon into or out of the reservoir system only occurs through the wells.

Note: The movement or location of the carbon after it leaves the reservoir system is also unimportant for tracking net carbon production/injection. It is unimportant if the produced carbon is reinjected (because it is included as injection at the injector well heads) or if the injected carbon comes from elsewhere.

Required input data (E100)

To activate the net carbon production/injection tracking feature in Eclipse 100, the 'carbon content' must be specified for either oil or gas (or both) with the [CARBOIL](#) and [CARBGAS](#) keywords. The carbon content is defined as the weight-moles of carbon per surface volume in the unit system used in the model, and a value can be supplied for each PVT region. As for other per-PVT-region keywords, if a region other than the first is defaulted, it takes the value from the previous region. The PVT region used in all calculations is the region of the producing or injecting well.

Carbon content cannot be specified for water.

Required input data (E300)

To activate the net carbon production/injection tracking feature in Eclipse 300, the 'carbon content' must be specified for any components that contain carbon with the [CARBON](#) keyword. The carbon content of a

component is defined as the number of moles of carbon per mole of the component. For example, one mole of propane (C_3H_8) contains three moles of carbon atoms so it has a carbon content value of 3.

Carbon content cannot be specified for water components.

Note: In E300, CARBON can be used for Black oil simulation. Internally, the black oil description is converted to an OIL and a GAS component. The first value will correspond to the OIL component and the next one to GAS. Note that the molecular weight of OIL is hardcoded to 120 and the molecular weight of GAS is hardcoded to 25.

Conversion from moles to equivalent mass

In the output at the end of the PRT file, in addition to the net carbon moles produced or injected, the 'equivalent mass of carbon-12' and the 'equivalent mass of CO_2 ' are also reported. These are calculated as follows:

Equivalent mass of carbon-12

*Equivalent mass of carbon-12 [lb] = (moles of carbon [lb-mole]) * (molecular weight of carbon-12 [lb/lb-mole])*

where the molecular weight of carbon-12 is 12 in all unit systems (by definition). The resulting value is then output in the appropriate unit for the unit system used in the model.

Equivalent mass of CO_2

Because each mole of CO_2 contains one mole of carbon atoms, the total amount of CO_2 produced by burning the net carbon molar production in the presence of sufficient oxygen is:

*Equivalent mass of CO_2 [lb] = (moles of carbon [lb-mole]) * (molecular weight of CO_2 [lb/lb-mole])*

where the molecular weight of CO_2 is 44.01 in all unit systems. As above, the resulting value is then output in the appropriate unit for the unit system used in the model.

Note: The 'equivalent mass of CO_2 ' reported here is different to the similarly named ' CO_2 equivalent', which is commonly used in carbon accounting literature. CO_2 equivalent is a measure of how much a gas contributes to global warming, relative to carbon dioxide, and is calculated based on the mass of the gas (in tonnes) multiplied by the published global warming potential (GWP) of the gas. Such a value does not easily lend itself to the reservoir simulation system where much of the carbon produced is in the liquid phase.

Reporting

For each well and each group, summary output for the time-varying values of the rate of carbon molar production (where negative indicates injection) and the cumulative molar carbon production during the simulation can be requested. For example:

```
WNCPR
  'P*' /
  'I*' /
/
FNCPT
```

The `FNCPT` value for the field at the end of the simulation corresponds to the 'moles' row in the table at the end of the PRT file.

Carbon reporting can be added in a restart.

Eclipse 300

The cumulative carbon produced will account for production / injection during the base case using the value provided in the restart case.